

The a_0 – H_0 relation in Λ -EG: a consistency test and its distance-ladder circularity

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Abstract

The Lambda Emergent Gravity (Λ -EG) framework fixes two acceleration scales in a rigid geometric relation: the screening scale $g_{\text{crit}} = cH_0/(4\pi^2\sqrt{3})$ derived from de Sitter spectral geometry [Figueroa Torres 2026], and the MOND acceleration $a_0 = 4\pi g_{\text{crit}}$ derived from the AQUAL embedding [Figueroa Torres 2026b, §4]. We write out the immediate cosmological consequence of these two identities, previously distributed across companion papers but never stated explicitly: $H_0 = \pi\sqrt{3}a_0/c$. Substituting the SPARC-canonical $a_0 = (1.20 \pm 0.02_{\text{stat}} \pm 0.24_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2}$ yields $H_0 = 67.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$, within 0.3% of the Planck 2018 CMB central value. We then show — and this is the main cautionary result of the letter — that this agreement cannot at present be read as an independent determination: between 55% and 74% of the standard SPARC sample (depending on how cluster-assigned distances are counted) carries distances computed under an assumed $H_0 = 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and because $a_0 \propto D^{-2}$ these propagate into a_0 with sample-weighted sensitivity $d \ln a_0 / d \ln H_0 = 1.1$ – 1.5 , exceeding unity and comparable to the tension being probed. Imposing the relation and the calibration simultaneously yields a loop that is singular at $H_0 \simeq 72$ and therefore determines no value of H_0 at all. We present the relation as a consistency condition, quantify the coupling explicitly, and identify the clean route: the 33 SPARC galaxies with individually measured redshift-independent distances. We then observe that the companion observable $\sigma_\Phi(z)$, the intrinsic scatter of the baryonic Tully–Fisher relation across redshift, measures the shape $E(z) \equiv H(z)/H_0$ independently of H_0 itself, since $\sigma_\Phi(z)/\sigma_\Phi(0) = E(z)^{\beta'(0)}$ [Figueroa Torres 2026b, §4 and Appendix A.11]. The two observables are informationally orthogonal: a_0 carries the scale, σ_Φ carries the shape. Combined, they reconstruct $H(z) = H_0 E(z)$ from galactic dynamics alone, using neither the CMB nor a distance ladder. We register predictions before ELT/JWST-IFU deployment. All algebraic results follow from previously published derivations; the new content is their combination and its empirical evaluation.

Keywords: Hubble tension; MOND; emergent gravity; baryonic Tully–Fisher relation; distance-ladder systematics

1 Introduction

The Hubble tension — the persistent, now $\gtrsim 5\sigma$ discrepancy between the value of H_0 inferred from the cosmic microwave background under Λ CDM [Planck Collaboration 2020] and the value measured directly through the Cepheid–supernova distance ladder [Riess et al. 2024] — has grown rather than shrunk with each generation of data. JWST observations of the Cepheid rungs have ruled out crowding and dust as the primary systematic in the local rung [Riess et al. 2024], and the two determinations remain approximately $6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ apart. The tension therefore now motivates either an unrecognized systematic in one of the two anchors, or a modification of the cosmological model between the recombination epoch and today.

A third measurement path, independent of both anchors, would clarify which possibility to take seriously. This letter identifies such a path within Λ -EG, using observables that live entirely on galactic scales and therefore share no calibration systematic with either the CMB or the distance ladder. The path splits naturally into two orthogonal components: a present-epoch determination of H_0 from the MOND acceleration scale a_0 , and a redshift-evolution determination of the shape $E(z) = H(z)/H_0$ from the intrinsic scatter of the baryonic Tully–Fisher relation (BTFR). Neither is new mathematics: the ingredients have all been derived and published in the two companion papers [Figueroa Torres 2026, 2026b]. What is new is their explicit combination, and its empirical evaluation.

We are honest at the outset about the strength of the present claim. The determination of H_0 from a_0 is currently limited by the $\sim 20\%$ systematic on a_0 from stellar-mass modelling and inclination corrections in SPARC [Lelli, McGaugh & Schombert 2016]; the resulting band on H_0 is wide enough to encompass both Planck and SH0ES. It is further limited, as we show in §4, by the fact that a majority of the SPARC distances are themselves computed under an assumed H_0 , so that the present-epoch channel is not yet ladder-independent in the strict sense. What Λ -EG delivers today is therefore neither a statistical arbitration of the tension nor a fully independent third measurement, but a structural relation between two acceleration scales that is testable, a quantified account of the circularity that currently limits it, and a program to remove that circularity through improved a_0 constraints and through ELT/JWST-IFU measurements of $\sigma_\Phi(z)$. The second channel, $\sigma_\Phi(z)$, is immune to the problem: being a ratio, it is independent of H_0 by construction.

2 The a_0 – H_0 relation

Λ -EG postulates two acceleration scales that are related by a fixed geometric factor. The screening scale g_{crit} was derived in the framework paper from de Sitter spectral geometry as

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12} \text{ m s}^{-2}, \quad H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (1)$$

[Figueroa Torres 2026, §3]. The AQUAL embedding [Bekenstein & Milgrom 1984] of the same framework then fixes the deep-MOND acceleration scale a_0 that enters the interpolating-function argument $x = g/a_0$ as

$$a_0 = 4\pi g_{\text{crit}}, \quad (2)$$

where the factor 4π is the solid-angle weight of the Gauss-theorem flux integration over a closed surface enclosing a spherically symmetric baryonic source in the deep-MOND regime [Figueroa Torres 2026b, §4]. Substituting (1) into (2) and solving for H_0 :

$$\boxed{H_0 = \frac{\pi\sqrt{3}a_0}{c}} \quad (3)$$

The prefactor $\pi\sqrt{3} \simeq 5.4414$ is a pure geometric constant. The $\sqrt{3}$ is the square root of the spectral gap of the Laplacian on the S^3 sections of de Sitter, $\lambda_1(S^3) = 3$, which the framework paper takes as the primary origin of the factor [Figueroa Torres 2026, §4.4]. The identity $\lambda_1(S^D) = D$ makes the count independent of D across round spheres, and the framework argues separately that the same $\sqrt{D}$ follows from counting D isotropic decoherence channels, which does not presuppose a spherical section. We avoid the stronger phrasing “topology-independent”: $\lambda_1 = D$ is a property of constant-positive-curvature space forms, not of arbitrary topologies — on a flat T^3 the gap depends on the torus moduli, on H^3 it is set by the curvature radius, and on $\mathbb{R}^3$ the spectrum is continuous with no gap at all. The claim that survives is the weaker one: within space forms of constant positive curvature the factor is fixed by the spatial dimension alone. The π comes from the closed-surface integration of (2).

Equation (3) is not a new derivation. It is the direct consequence, in a single line of algebra, of (1) and (2), both previously published. Its cosmological content lies in the recognition that a_0 is measured on galactic scales, through observables (rotation velocities, photometric masses) that involve no cosmological model, so that (3) delivers H_0 without invoking the CMB or the distance ladder *as theoretical inputs*. Whether the distances used to convert those observables into physical units are themselves free of H_0 is a separate question, and §4 shows that at present they are not.

Two remarks. First, the factor 4π in (2) is not a phenomenological fit to SPARC: it is the Gauss-theorem projection weight, structural to any scalar-field theory whose deep-MOND behaviour reduces to AQUAL. The observed agreement with SPARC at the $\sim 4\%$ level is therefore data-limited rather than fit-limited. Second, (3) is a hard prediction of the combined Λ -EG + AQUAL structure: it cannot be relaxed by choosing a different interpolating function, since g_{crit} and a_0 are both defined outside the interpolant, at the asymptotic Newtonian and deep-MOND boundaries respectively.

3 Empirical evaluation

The SPARC survey [Lelli, McGaugh & Schombert 2016; McGaugh, Lelli & Schombert 2016] measures a_0 from the radial-acceleration relation across approximately 150 disk galaxies, obtaining the canonical value

$$a_0 = (1.20 \pm 0.02_{\text{stat}} \pm 0.24_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2}, \quad (4)$$

where the systematic uncertainty is dominated by stellar mass-to-light modelling and inclination corrections. Substituting the central value into (3),

$$H_0 = \frac{\pi\sqrt{3} \times 1.20 \times 10^{-10}}{2.998 \times 10^8} \text{ s}^{-1} = 67.21 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (5)$$

Propagating the SPARC systematic on a_0 directly through (3) gives the interval $H_0 \in [53.8, 80.6] \text{ km s}^{-1} \text{ Mpc}^{-1}$. Both reference determinations sit comfortably inside it: Planck (67.4 ± 0.5) at $0.01\sigma_{\text{sys}}$ and SH0ES (73.5 ± 0.7) at $0.47\sigma_{\text{sys}}$ from the central value. We stress that the band does not discriminate between them at all — the proximity of the central value to Planck carries no statistical weight on its own. The comparison is summarised in Table 1.

Method	H_0 [$\text{km s}^{-1} \text{ Mpc}^{-1}$]	Anchor
Planck 2018 CMB	67.4 ± 0.5	recombination-epoch acoustic scale
SH0ES 2024	73.5 ± 0.7	Cepheid–SN Ia distance ladder
This letter (Eq. 3)	67.2 (central)	galactic a_0 via SPARC
Δ vs. Planck	-0.3% (central)	
Δ vs. SH0ES	-8.6% (central)	

Table 1: Comparison of H_0 determinations from three independent anchors. The Λ -EG prediction (this letter) uses no CMB data and no distance ladder; its input is the SPARC-canonical value of a_0 . The wide statistical band from the SPARC systematic ($\pm 20\%$ on a_0 translates to $\pm 13.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ on H_0) prevents statistical arbitration at this data quality; the coincidence of central values with Planck is nevertheless notable given the total independence of the two anchors.

The central value coincides with the Planck determination at a fractional precision of 3×10^{-3} ; the same central value differs from SH0ES by 8.6%. Given that a shift of the Λ -EG central value from 67.21 to the SH0ES central of 73.5 would require a_0 to have a true value of $\sim 1.31 \times$

$10^{-10} \text{ m s}^{-2}$ (a 9% excess over the SPARC central), and given the empirical precision expected from Gaia-anchored extended SPARC and LSST-era radial-acceleration-relation determinations [order-of-magnitude reduction in the a_0 systematic on a decade timescale], Eq. (3) has the potential to become a sharp test. That potential is conditional on removing the circularity quantified in §4: a tighter a_0 derived from the same distance mix would be a tighter number, not a more independent one.

4 The H_0 content of the a_0 measurement

The argument of §3 is only as independent of the distance ladder as the distances that enter a_0 . We examined this directly and found that it is not independent at present data quality. The result is reported here rather than in a footnote because it bounds the claim the letter can make.

Table 1 of SPARC [Lelli, McGaugh & Schombert 2016] carries a distance-method flag f_D whose first value is defined in the machine-readable header as “*Hubble-Flow assuming $H_0 = 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and correcting for Virgo-centric infall*”. All counts below are on the standard BTFR sample — quality flag $Q \leq 2$ with a measured V_{flat} , which is 129 of the 175 galaxies in the table, not on the full 175.

Not every non-flow method is equally free of H_0 , and the distinction matters enough to be made explicit (Table 2). We separate three tiers. *Primary* distances (TRGB, Cepheids, SNe Ia; 33 galaxies) are measured to the individual galaxy by a stellar or standard-candle indicator: these are redshift-independent in the NED-D sense. They are not free of the distance-ladder *zero point* — TRGB and Cepheid calibrations share anchors with SH0ES (NGC 4258, LMC eclipsing binaries, Gaia parallaxes) — but no value of H_0 enters the inference. The *cluster* tier (Ursa Major, 25 galaxies) is ambiguous: SPARC assigns all 25 members the *same* distance, 18.0 Mpc, so it is a membership assignment rather than an individual measurement, and the cluster distance itself rests on a Tully–Fisher calibration that carries a distance-scale dependence. The *flow* tier (71 galaxies) is unambiguously H_0 -dependent.

Tier	Method	N	fraction
flow	Hubble flow ($D = v/H_0$ at $H_0 = 73$)	71	55.0%
cluster	Ursa Major (all assigned 18.0 Mpc)	25	19.4%
primary	Tip of the red giant branch	28	21.7%
primary	Cepheids	3	2.3%
primary	Supernovae Ia	2	1.6%
f (loose: flow only)		71	55.0%
f (strict: flow + cluster)		96	74.4%
genuinely redshift-independent		33	25.6%

Table 2: Distance provenance of the standard SPARC BTFR sample (129 galaxies). Depending on how the Ursa Major cluster tier is counted, between 55% and 74% of the sample carries distances that assume a value of H_0 — and that assumed value (73) sits at the SH0ES end of the tension. Only 33 galaxies have individually measured redshift-independent distances.

We carry both accountings forward, writing f for the H_0 -dependent fraction: $f = 0.550$ (loose) or $f = 0.744$ (strict). Nothing below depends on which is adopted.

For a galaxy whose distance comes from the Hubble flow, $D \propto 1/H_0$. Baryonic masses are inferred photometrically, so $M \propto D^2$, while V_{flat} is a velocity and carries no distance dependence. Since $a_0 = V_{\text{flat}}^4/(GM)$ in the BTFR normalisation,

$$a_0 \propto \frac{1}{M} \propto \frac{1}{D^2} \propto \left(H_0^{\text{assumed}}\right)^2 \quad \text{on the } f_D = 1 \text{ sub-sample,} \quad (6)$$

The exponent in (6) is exact rather than estimated, and it is worth seeing why. In the deep-MOND regime the interpolant gives $a^2 = a_0 a_N$, so that $V^4/r^2 = a_0 GM/r^2$ and *the radius cancels identically*: the BTFR contains no length scale. The distance therefore enters through the baryonic mass alone — V_{flat} being a Doppler velocity that carries no distance dependence — and does so with the single power $M \propto D^2$. Rescaling every distance in the sample by a common factor λ and recomputing returns $a_0 \propto \lambda^{-2}$ to five decimal places, as it must.

A second structural observation sharpens what this does and does not indict. In $M = V^4/(Ga_0)$ the powers of time cancel exactly ($[V^4] = \text{m}^4 \text{s}^{-4}$, $[Ga_0] = \text{m}^4 \text{kg}^{-1} \text{s}^{-4}$), so the BTFR carries no net time dimension: H_0 , an inverse time, cannot enter the relation *structurally*. It can enter only through the calibration of the data — the conversion of fluxes into masses, which requires a distance. The circularity documented in this section is therefore a property of the present SPARC distance mix, not of Eq. (3), and that is why the remedy is a better-selected sample rather than abandonment of the relation.

Propagating (6) gives a sample-weighted sensitivity $d \ln a_0 / d \ln H_0 = 2f$, that is 1.10 (loose) or 1.49 (strict). Either way the coefficient exceeds unity: a 1% error in the assumed H_0 moves a_0 by more than 1%. A shift of the assumed H_0 across the Planck–SH0ES range ($73 \rightarrow 67.4$, i.e. -7.7%) moves a_0 by -8% to -11% : the same size as the tension the relation is meant to probe.

The consequence is that Eq. (3) and the calibration of a_0 are not independent equations. Imposing both simultaneously,

$$H_0 = K g(H_0), \quad g(H) \equiv f \left(\frac{H}{73} \right)^2 + (1 - f), \quad K \equiv \frac{\pi \sqrt{3} a_0^{\text{SPARC}}}{c} = 67.21, \quad (7)$$

gives roots at $H_0 = 43.1$ and $100.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for $f = 0.550$, and at 21.6 and 85.0 for $f = 0.744$. None of the four is 67.2 . The value quoted in §3 is the solution of (3) at *fixed* a_0 — that is, holding constant the very quantity that was calibrated against $H_0 = 73$.

The conditioning of (7) is more informative than its roots. Differentiating, the response of the inferred H_0 to the published a_0 is

$$\frac{d \ln H_0}{d \ln a_0} = \frac{K}{H_0} \frac{g(H_0)}{1 - K g'(H_0)}, \quad g'(H) = \frac{2fH}{73^2}. \quad (8)$$

For $f = 0.550$ the denominator vanishes at $H_0 = 72.0$ — *inside* the range under dispute. The response is $+13.7$ at $H_0 = 67.2$, $+32.5$ at 70 , and changes sign to -68.5 by 73 : the loop is singular precisely where the tension lives, and determines nothing. For $f = 0.744$ the singularity moves out to 53.3 and the response is a finite -3.4 to -2.5 — still large, and still *negative*, meaning a higher published a_0 implies a *lower* self-consistent H_0 , the opposite of the naive reading of (3).

Under either accounting the loop fails to determine H_0 , and that failure — not any particular root — is the result of this section. We do not present (7) as a competing determination: it uses a linear sample-weighted propagation, whereas the published a_0 comes from a global fit to the radial acceleration relation rather than from a median over Table 1, and the Virgo-centric infall correction is not a rigid rescaling. The roots are an order-of-magnitude statement about the size of the coupling, and that size is large enough to void the independence claim.

We also attempted the direct test: recomputing a_0 separately on the two sub-samples. The 58 galaxies with H_0 -independent distances give a median a_0 that is $+9.5\%$ higher than the 71 Hubble-flow galaxies. However, binning by V_{flat} shows the offset does not hold a consistent sign ($+43\%$, $+18\%$, -11% , $+1\%$ from the lowest to the highest mass bin), so it is degenerate with mass-dependent selection between the sub-samples and is *not* a clean measurement of the H_0 bias. We report it as a bound on what Table 1 alone can resolve, not as a detection.

The clean route is available and we state it as the next step rather than attempt it here: re-derive a_0 from the 58 galaxies with H_0 -independent distances, using the full rotation curves and mass modelling rather than the summary table. That determination would be genuinely

ladder-independent, and Eq. (3) applied to it would be the measurement this letter currently cannot make.

5 The orthogonal companion: $\sigma_\Phi(z)$ as a probe of $E(z)$

The BTFR flow derived from the interpolant $\mu(x) = x/\sqrt{1+x^2}$ [Figueroa Torres 2026b, §4] produces the intrinsic-scatter evolution

$$\sigma_\Phi(z) = \sigma_\Phi(0) \left(\frac{H(z)}{H_0} \right)^{\beta'(0)} = \sigma_\Phi(0) E(z)^{\beta'(0)}, \quad (9)$$

where $E(z) = H(z)/H_0 = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ and $\beta'(0) = -1$ for the standard interpolant of the theorem. The key observation is that $E(z)$ depends only on the density parameters Ω_m, Ω_Λ and *not* on H_0 itself: the identical $E(2) = 3.032$ is obtained for any $H_0 \in [65, 75] \text{ km s}^{-1} \text{ Mpc}^{-1}$ holding $\Omega_m = 0.315$ fixed. Equation (9) therefore measures the *shape* of the expansion history, not its overall scale.

Symmetry protection: why (9) is well-defined

The reconstruction (9) is not automatic. If the interpolating function’s Taylor expansion at the deep-MOND fixed point had a non-vanishing quadratic coefficient c_2 , the flow would become non-Lipschitz and $\sigma_\Phi(z)$ would collapse to zero exactly at finite flow time [Figueroa Torres 2026b, §5], destroying the smooth $E(z)^{\beta'(0)}$ dependence on which the reconstruction relies. The Appendix A.11 of that paper formalises this: $c_2 = 0$ is a knife-edge condition requiring symmetry protection, and the isotropy axiom of the theorem [Figueroa Torres 2026b, §2] provides it exactly. The condition is moreover not imposed by hand: among the four simple projection gates whose solid-angle fractions generate the candidate interpolants, only the tilt gate $g \sin \theta > a_0 \cos \theta$ yields a μ with vanishing quadratic coefficient [Figueroa Torres 2026b, §3]. The reconstruction (9) therefore rests on a symmetry-protected structural identity with a geometric origin, not on a phenomenological choice. Any competing MOND-like framework whose interpolant does not enforce $c_2 = 0$ will not yield a smooth $\sigma_\Phi(z)$ curve, and its inability to do so will be visible in first-generation ELT/JWST-IFU measurements at $z \gtrsim 1$.

Orthogonality with a_0

The two observables carry orthogonal information about the expansion history:

Observable	Anchor epoch	What it measures
a_0 (SPARC)	$z \simeq 0$	H_0 (scale) via Eq. (3)
$\sigma_\Phi(z)$ (ELT/JWST-IFU)	$z \simeq 1\text{--}2$	$E(z) = H(z)/H_0$ (shape) via Eq. (9)

Combined, they reconstruct $H(z) = H_0 \cdot E(z)$ across the redshift range accessible to resolved kinematics, without invoking the CMB or the distance ladder at any step. The reconstruction is closed even without external priors on Ω_m : two measurements of σ_Φ at distinct redshifts fix both the shape parameter and, through $E(z_1)/E(z_2)$, provide a consistency check against the assumed background cosmology.

6 Combined reconstruction and registered predictions

Table 3 lists the predictions of $\sigma_\Phi(z)$ for the three surviving interpolants of [Figueroa Torres 2026b, Table 1], under two cosmological scenarios: the Planck-consistent Λ CDM background ($\Omega_m = 0.315, H_0 = 67.4$) and a SH0ES-consistent variant ($\Omega_m = 0.30, H_0 = 73.5$). Because

$E(z)$ is invariant under the choice of H_0 at fixed Ω_m , the two scenarios differ only through the modest ~ 0.015 shift in Ω_m typically accompanying the higher- H_0 solutions. We anchor $\sigma_\Phi(0) = 0.07$ dex following the observed SPARC intrinsic scatter [Lelli et al. 2019].

Interpolant	$\beta'(0)$	$\sigma_\Phi(z=0.5)$	$\sigma_\Phi(z=1.0)$	$\sigma_\Phi(z=2.0)$	$\sigma_\Phi(z=4.0)$
<i>Planck-consistent</i> ($\Omega_m = 0.315$, $H_0 = 67.4$):					
Standard $\mu = x/\sqrt{1+x^2}$	-1	0.053	0.039	0.023	0.011
$\hat{\mu}$ entropic	-2	0.040	0.022	0.008	0.002
$\tanh(x)$	-2/3	0.058	0.047	0.033	0.020
<i>SH0ES-consistent</i> ($\Omega_m = 0.30$, $H_0 = 73.5$):					
Standard $\mu = x/\sqrt{1+x^2}$	-1	0.053	0.040	0.024	0.011
$\hat{\mu}$ entropic	-2	0.041	0.023	0.008	0.002
$\tanh(x)$	-2/3	0.059	0.048	0.034	0.021

Table 3: Registered predictions of $\sigma_\Phi(z)$ [dex] under two cosmological scenarios. The insensitivity to H_0 at fixed Ω_m (compare rows across the two blocks) is the orthogonality property of §5. The dominant discriminator is $\beta'(0)$: for a target error budget of ± 0.010 dex per redshift bin, the standard interpolant is separable from $\hat{\mu}$ near $z \simeq 1.2$ and from $\tanh$ near $z \simeq 2.3$ [Figuroa Torres 2026b, Fig. 1]. Values computed at fixed $\sigma_\Phi(0) = 0.07$ dex. Registered before ELT/JWST-IFU deployment.

The insensitivity of $\sigma_\Phi(z)$ predictions to the choice of H_0 under fixed Ω_m is the direct expression of the orthogonality established in §5: the shape measurement does not carry scale information, and vice versa. What the shape measurement *does* carry is discriminating power between models that resolve the Hubble tension through modifications of $E(z)$. Early-dark-energy models, for instance, modify $E(z)$ at $z \gtrsim 1000$ but leave the late-time expansion essentially untouched; models that modify late-time dark-energy dynamics predict deviations at $z \lesssim 1$. A measurement of $\sigma_\Phi(z = 1.5)$ at ± 0.005 dex precision — within reach of ELT/HARMONI resolved kinematics with SED-fit baryonic masses — would separate these classes at the $\gtrsim 3\sigma$ level, independently of any determination of the CMB acoustic scale.

7 Epistemological disclosure

We are explicit about what has been fit, what has been derived, and what remains open.

Fit to data: none of the numbers in this letter. The prefactor $\pi\sqrt{3}$ in Eq. (3) is a pure geometric constant; the factor 4π in Eq. (2) is the Gauss-theorem projection weight, structural to the AQUAL embedding and independent of the choice of interpolating function. The value of a_0 entering Eq. (5) is the SPARC-canonical determination. It is *not*, however, independent of H_0 : as quantified in §4, 55% of the sample carries Hubble-flow distances computed at $H_0 = 73$, coupling a_0 to the assumed expansion rate with sensitivity $d \ln a_0 / d \ln H_0 \simeq 1.1$. An earlier version of this letter asserted that independence; the assertion was wrong and is retracted here.

Branch ambiguity: Eq. (3) presupposes that the acceleration scale tracks the instantaneous expansion rate, $a_0 \propto H(z)$. This is one of three readings of the framework’s scale relation. The covariant reading $a_0 \propto \sqrt{R(z)}$ is excluded independently of the present data, since $R = 8\pi G(\rho - 3p)$ vanishes exactly in the radiation era while the expansion $\theta = 3H$ does not [Figuroa Torres 2026, §7.6]. The third reading, $a_0 \propto \sqrt{\Lambda}$ (constant in z), is not excluded, and it would make Λ -EG indistinguishable from MOND in redshift.

Each branch predicts a different a_0 at fixed H_0 , so each inverts to a different H_0 at fixed a_0 . Evaluated at $H_0 = 67.4$ the three predictions are $a_0^A = 1.203$, $a_0^B = 1.051$ and $a_0^C = 0.995$, in units of $10^{-10} \text{ m s}^{-2}$ [Figuroa Torres 2026, §7.6]. Since each branch has $a_0 \propto H_0$ at fixed Ω_Λ ,

inverting on the observed $a_0 = 1.20$ gives

$$H_0^{\text{branch}} = 67.4 \times \frac{1.20}{a_0^{\text{branch}}/10^{-10}} = 67.2 \text{ (A)}, \quad 77.0 \text{ (B)}, \quad 81.3 \text{ (C)}, \quad (10)$$

an ambiguity of 21%, larger than the 8.4% tension being probed. The discrepancy of each branch is properly quoted in a_0 , where the 20% systematic lives: branch C misses the central value by $(1.20 - 0.995)/0.24 = 0.85\sigma_{\text{sys}}$ and branch B by $0.62\sigma_{\text{sys}}$. (Quoting branch C’s offset against the *branch-A* band of ± 13.4 would give a different and incorrect 1.05σ : that band is $\pi\sqrt{3}\sigma_{a_0}/c$, the propagation of σ_{a_0} through branch A’s constant, and does not apply to a displacement generated by a different branch.) We note that $0.85\sigma_{\text{sys}}$ is weak grounds for exclusion. The reader should treat 67.2 as conditional on the $a_0 \propto H(z)$ branch.

We stress that (10) is a different calculation from the self-consistency roots of §4 and the two should not be conflated: the former asks which H_0 reproduces the observed a_0 under each scale relation, the latter asks which H_0 is consistent with a dataset calibrated at $H_0 = 73$. Setting $f = 1$ in (7), for instance, returns $73^2/K = 79.3$, which is neither 81.3 nor a branch statement at all.

Structural inputs: the framework paper adopts the AQUAL embedding as its scalar sector; the derivation of AQUAL from a more fundamental principle within the covariant action of Λ -EG remains as Open Problem 6 [Figueroa Torres 2026, §13]. This does not affect the present letter, since Eq. (2) holds within AQUAL as a mathematical consequence, but it means the ultimate justification of Eq. (3) inherits whatever justification AQUAL receives from future work.

Observational inputs: the SPARC value of a_0 carries a $\sim 20\%$ systematic uncertainty at present. This uncertainty, propagated linearly through Eq. (3), produces the wide $\pm 13.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ band on H_0 . Statistical arbitration of the Hubble tension will therefore require the SPARC systematic to be reduced to $\lesssim 5\%$; the pathway is through improved stellar-mass modelling (Gaia parallax priors), tighter inclination priors from resolved kinematics, and larger samples (extended SPARC, LSST-era radial-acceleration relation catalogues). Reducing the systematic is necessary but not sufficient: the distance provenance of §4 must be addressed separately, since it is a bias rather than a variance and does not shrink with sample size.

Ex-ante predictions: the numerical entries of Table 3, registered on the deposition date of this letter, will be compared against ELT/HARMONI IFU measurements of resolved BTFR kinematics at $z \simeq 1\text{--}3$ when they become available. Should the observed $\sigma_\Phi(z)$ deviate from the standard-interpolant prediction under either cosmological scenario, one or more of the following will be falsified: the isotropy axiom of the theorem, the AQUAL embedding, the Λ CDM background, or the assumption that a_0 is redshift-invariant in the observer frame.

What we do not claim: we do not claim to have resolved the Hubble tension, and we do not claim a third independent determination of H_0 . Two things stand in the way, and we have quantified both: the width of the SPARC systematic on a_0 precludes statistical arbitration at the level demanded by the 5σ CMB–distance-ladder discrepancy, and the distance provenance of the sample (§4) prevents the present-epoch channel from being ladder-independent at all; the second is a bias, and §4 shows the resulting loop is ill-conditioned in exactly the range under dispute. We claim, more modestly: that Λ -EG fixes a rigid algebraic relation between a_0 and H_0 with no free parameters; that evaluating it on current data returns a value 0.3% from Planck, which is suggestive but cannot be counted as evidence until the circularity is removed; that the route to removing it is concrete and identified (the 33 galaxies with individually measured redshift-independent distances); and that the orthogonal companion (§5), being a ratio and therefore immune to the distance problem, opens a program to reconstruct the *shape* of $H(z)$ from galactic dynamics alone.

8 Discussion and outlook

The a_0 – H_0 coincidence has been noticed in the MOND literature since at least Milgrom (1999), who observed that $a_0 \sim cH_0$ in order-of-magnitude terms and interpreted it through the de Sitter Unruh temperature. Milgrom (2009) sharpened the connection considerably, deriving the deep-MOND limit from space-time scale invariance and conjecturing that local gravitational physics would take exactly the deep-MOND form in an exact de Sitter universe, on the grounds that the isometry group of dS_4 and the symmetry group of the deep-MOND field equation are both $SO(4,1)$. The present letter is agnostic on that mechanism: Eq. (3) is used here only as an algebraic relation between two measured scales. Klinkhamer & Kopp (2011) derived the exact relation $a_0 = 4\pi(c/\hbar)k_B T_{\min}$ from entropic gravity with a minimum temperature. The present letter differs from those in two respects. First, the prefactor $\pi\sqrt{3}$ in Eq. (3) is derived, not observed: it follows from the specific de Sitter spectral geometry of the framework paper [Figueroa Torres 2026] combined with the AQUAL embedding. Second, and more importantly, the coincidence is complemented here by the orthogonal $\sigma_\Phi(z)$ observable, which promotes a numerical coincidence into a two-channel reconstruction program.

The path forward is observational. Four developments would sharpen this letter’s contribution to the tension debate, of which the first is prerequisite to the rest: (0) re-derivation of a_0 restricted to galaxies with redshift-independent distances — 33 have individually measured ones in SPARC, 58 if the Ursa Major cluster tier is admitted (§4) — which is what would convert Eq. (3) from a consistency relation into a measurement; (i) reduction of the SPARC systematic on a_0 to $\lesssim 5\%$ through Gaia-anchored mass modelling on extended galaxy samples; (ii) first-epoch measurements of $\sigma_\Phi(z)$ at $z \simeq 1$ –2 with ELT/HARMONI resolved kinematics, tested against the entries of Table 3; (iii) independent theoretical work on the AQUAL sector of the framework, either sharpening or falsifying the derivation of (2) from more fundamental principles.

We stress the falsifiability of the path. Eq. (3) is a rigid prediction: any determination of a_0 on H_0 -independent distances that lies more than $\sim 3\sigma$ below (4) forces Λ -EG into tension with Planck; any such determination more than $\sim 3\sigma$ above forces it into tension with SH0ES. The qualifier is essential — on the present distance mix the test is partly self-referential, and tightening the error bars alone would not make it decisive. Neither outcome is currently accessible with SPARC data quality, but both will be within a decade. Eq. (9) carries no such caveat, being a ratio in which H_0 cancels identically: any observed $\sigma_\Phi(z = 1.5) \notin [0.012, 0.040]$ dex (the union of the surviving-interpolant predictions of Table 3, computed at $\sigma_\Phi(0) = 0.07$ dex from SPARC: $\hat{\mu}$ -entropic 0.0125, standard 0.0296, tanh 0.0394) falsifies the combined framework.

Data availability

All numerical evaluations are reproducible from the archived scripts `H0_from_a0.py` (Eq. 3 and Table 1), `sigmaPhi_predictions.py` (Table 3) and `a0_H0_selfconsistency.py` (§4, Table 2 and Eq. 7). All three are deterministic and require only the Python standard library; the third additionally reads the machine-readable SPARC Table 1 (`SPARC_Lelli2016c.mrt`), archived alongside them. The symbolic verification of the fixed-point structure underlying Eq. (9) is in `sensitivity.py` (Appendix A.11 of the companion paper, reproduced here), which requires SymPy. All symbolic identities used in §2 and §5 have been verified in the archived scripts of the companion papers [Figueroa Torres 2026, 2026b].

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